Parameter Estimation

Likelihood, maximum likelihood, and least squares

Kangning Cui
City University of Hong Kong (Dongguan)

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Course Project: Purpose and Expectations

The course project is an opportunity to apply machine-learning principles to a real problem. A strong project should include:

- a clearly defined problem;
- an appropriate dataset;
- a justified choice of methods;
- meaningful evaluation and comparison;
- analysis of results and limitations.

The goal is to understand why it works and how to evaluate it.

Project team size and contribution

The recommended team size is

$$\boxed{3\text{–}6 \text{ students}}$$

However, an individual project is also acceptable if the scope is appropriate and the student can complete the work independently.

Contribution requirement

Every project must clearly describe:

- who contributed to each part;
- what each member completed;
- how the work was divided.

Project quality and contribution matter more than team size.

Choosing a project topic

Good project directions include:

- **Remote sensing**
  e.g., land-cover classification, object detection, environmental monitoring.
- **Medical AI**
  e.g., medical image analysis, clinical prediction, signal analysis.
- **Large language models**
  e.g., evaluation, adaptation, retrieval-augmented systems.
- Other machine-learning problems with a clear dataset and evaluation plan.

Choose a problem where data, methods, and evaluation can be clearly justified.

How to find a suitable project

1. **Find a dataset**
   - Kaggle;
   - Hugging Face Datasets;
   - public research datasets.
2. **Understand existing methods**
   Search recent papers from:
   - NeurIPS, ICML, ICLR;
   - CVPR, ICCV, ECCV;
   - ACL, EMNLP, NAACL;
   - relevant journals.
3. **Define your contribution**
   Ask:
   - What problem are we solving?
   - What baseline will we compare with?
   - How will success be measured?

Start early and discuss your ideas

Recommended timeline:

- Weeks 1–4: explore topics and datasets;
- Around Week 5: finalize the problem and experimental plan;
- Later weeks: implement, compare, analyze, and improve.

Need feedback?

If you are unsure whether a topic is suitable, please email the instructor with:

- problem description;
- dataset;
- expected methods;
- evaluation plan.

Learning objectives

From probability models to parameters learned from data

By the end of this lecture, you should be able to:

1. explain the difference between probability and likelihood;
2. construct likelihood functions from observed data;
3. derive maximum-likelihood estimates for simple models;
4. understand why more data usually leads to more stable estimates;
5. connect probabilistic assumptions with loss functions.

From a model to an objective

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The parameter-estimation problem

A probability model describes how observations are generated. Some quantities in the model are unknown and need to be estimated from data.

Example: coin tossing

Let $x = 1$ represent heads and $x = 0$ represent tails. The probability of heads is an unknown parameter π:

$$p(x \mid \pi) = \pi^x (1 - \pi)^{1-x}.$$

Given observations

$$\mathcal{D} = \{x_1, \ldots, x_n\},$$

parameter estimation asks:

How should we estimate π from the observed data?

Probability versus likelihood

The expression $p(x \mid \theta)$ can answer two different questions.

Probability

The parameter is fixed and observations vary:

$$\theta \text{ fixed}, \qquad x \text{ varies}.$$

$p(x \mid \theta)$ is a probability distribution over possible observations:

$$\sum_{x} p(x \mid \theta) = 1.$$

Likelihood

The observations are fixed and parameters vary:

$$L(\theta; \mathcal{D}) = p(\mathcal{D} \mid \theta).$$

Likelihood compares parameter values. It is not a probability distribution over θ, so it does not need to sum to one.

Example: after observing 8 heads in 10 tosses, the data stay fixed while different values of π are compared.

Independent observations make the likelihood a product

We often assume observations are **i.i.d.**: observations are independent and follow the same distribution. Independence allows the joint likelihood to factorize:

$$p(\mathcal{D} \mid \theta) = \prod_{i=1}^{n} p(x_i \mid \theta).$$

Since the data are observed, maximum likelihood chooses the parameter that makes the observed data most plausible:

$$\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta \in \Theta} p(\mathcal{D} \mid \theta).$$

Taking logarithms gives the equivalent objective

$$\ell(\theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta), \qquad \hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} \ell(\theta).$$

How do we maximize the likelihood?

Maximum likelihood turns estimation into an optimization problem:

$$\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} \ell(\theta).$$

To find the optimum, we usually:

1. Find candidate points where the slope is zero:
$$\frac{d\ell}{d\theta} = 0.$$
2. Check whether the candidate is a maximum:
$$\frac{d^2\ell}{d\theta^2} < 0.$$
3. Check constraints on the parameter.

Common mistake

A zero derivative only gives a candidate solution; it does not guarantee a maximum.

Beyond the derivative: what should we check?

Finding a zero derivative is only the first step.

Is it a maximum?

A stationary point may be:

- a maximum;
- a minimum;
- a flat region.

The curvature helps identify the type of point.

Is it allowed?

Parameters often have constraints.
Example:

$$0 \leq \pi \leq 1$$

The optimum may occur at the boundary.

A valid MLE solution must satisfy both the optimization condition and the parameter constraints.

Bernoulli maximum likelihood

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Example: estimating a coin probability

Suppose a coin is flipped n times.
Let $x_i = 1$ represent heads and $x_i = 0$ represent tails. The unknown probability of heads is π:

$$p(x_i \mid \pi) = \pi^{x_i}(1-\pi)^{1-x_i}, \qquad 0 \le \pi \le 1.$$

Given observations

$$\mathcal{D} = \{x_1, \ldots, x_n\},$$

the likelihood is

$$L(\pi) = \prod_{i=1}^{n} p(x_i \mid \pi).$$

Maximum likelihood asks: which value of π makes the observed flips most plausible?

Bernoulli MLE: from likelihood to log-likelihood

Let

$$m = \sum_{i=1}^{n} x_i$$

be the number of observed heads.
The likelihood becomes

$$L(\pi) = \pi^m (1-\pi)^{n-m}.$$

Maximizing the likelihood is equivalent to maximizing the log-likelihood:

$$\ell(\pi) = m \log \pi + (n-m) \log(1-\pi).$$

Bernoulli MLE: solve and interpret

Differentiate the log-likelihood:

$$\frac{d\ell}{d\pi} = \frac{m}{\pi} - \frac{n-m}{1-\pi}.$$

Setting the derivative to zero:

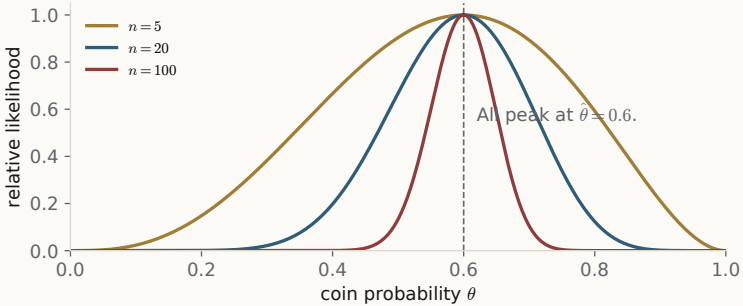
$$m(1-\pi) - (n-m)\pi = 0,$$

gives

$$\boxed{\hat{\pi}_{\mathrm{ML}} = \frac{m}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i}.$$

The MLE is simply the observed fraction of heads.

More data make the likelihood more concentrated



- The maximum remains near the observed fraction.
- More observations produce a sharper likelihood curve.
- More data give a more stable parameter estimate.

Gaussian maximum likelihood

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Example: estimating the mean of Gaussian data

Assume observations are independent and follow

$$x_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2),$$

where we first treat σ^2 as fixed and estimate the unknown mean μ.
The Gaussian density is

$$p(x_i \,|\, \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

For independent observations, the likelihood is

$$L(\mu) = \prod_{i=1}^{n} p(x_i \,|\, \mu).$$

Taking logarithms gives

$$\ell(\mu) = -\frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2 + \text{constant}.$$

Gaussian mean MLE

Differentiate the log-likelihood:

$$\begin{aligned}\frac{d\ell}{d\mu} &= -\frac{1}{2\sigma^2}\sum_i 2(x_i-\mu)(-1) \\ &= \frac{1}{\sigma^2}\sum_i (x_i-\mu).\end{aligned}$$

Setting this to zero gives

$$\sum_i x_i - n\hat{\mu} = 0 \quad \Longrightarrow \quad \boxed{\hat{\mu}_{\mathrm{ML}} = \bar{x}.}$$

The second derivative is $-n/\sigma^2 < 0$.

Example: estimating Gaussian variance

After estimating the mean $\hat{\mu} = \bar{x}$, we still need to estimate the variance.
The variance describes how far observations spread around the fitted mean:

$$\sigma^2 = \mathbb{E}[(X - \mu)^2].$$

For Gaussian data, maximum likelihood gives

$$\boxed{\hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{n} \sum_i (x_i - \bar{x})^2}.$$

This is the average squared distance from the estimated mean.

Common mistake

Maximum likelihood uses denominator n; unbiased estimation uses a different correction.

Why does estimating the mean change the variance?

Suppose we have three observations: $x_1,\ x_2,\ x_3$; if the true mean μ is known, deviations are measured from a fixed point:

$$(x_1 - \mu,\ x_2 - \mu,\ x_3 - \mu).$$

However, when μ is estimated from the same data,

$$\bar{x} = \frac{x_1 + x_2 + x_3}{3},$$

the residuals must satisfy

$$\sum_i (x_i - \bar{x}) = 0.$$

Once the mean is estimated, the residuals are no longer free to vary in all directions. One degree of freedom is used to estimate the mean.

Why does variance use $n-1$?

The maximum-likelihood estimate is $\hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{n}\sum_i (x_i - \bar{x})^2$.
Because one degree of freedom is used to estimate $\bar{x}$,

$$\mathbb{E}[\hat{\sigma}^2_{\mathrm{ML}}] = \frac{n-1}{n}\sigma^2.$$

Therefore, the MLE variance is slightly smaller than the true variance on average.
The unbiased sample variance corrects this effect:

$$S^2 = \frac{1}{n-1}\sum_i (x_i - \bar{x})^2.$$

From one Gaussian to multiple features

A one-dimensional Gaussian describes one variable:

$$\mathcal{N}(\mu, \sigma^2).$$

For multiple features,

$$\boldsymbol{x} = [x_1, \ldots, x_d]^\mathsf{T},$$

we need to describe:

- where the data are centered;
- how each feature varies;
- how different features move together.

Therefore, the variance becomes a covariance matrix:

$$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

Estimating a multivariate Gaussian

Given samples $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$, we estimate the center using the sample mean:

$$\hat{\boldsymbol{\mu}}_{\mathrm{ML}} = \frac{1}{n} \sum_i \boldsymbol{x}_i.$$

The covariance estimate is

$$\hat{\boldsymbol{\Sigma}}_{\mathrm{ML}} = \frac{1}{n} \sum_i (\boldsymbol{x}_i - \hat{\boldsymbol{\mu}})(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}})^{\mathsf{T}}.$$

Each entry has a meaning:

- diagonal: variance of one feature;
- off-diagonal: covariance between two features.

Estimating a multivariate Gaussian means estimating both the location and the relationships between features.

Estimator quality

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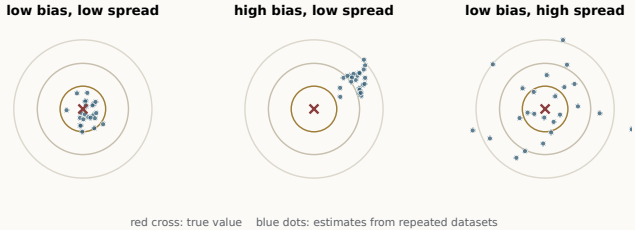
Why should we care about estimator quality?

The same estimation method can produce different results when we collect different data.

$$\mathcal{D}_1 \to \hat{\theta}_1, \qquad \mathcal{D}_2 \to \hat{\theta}_2, \qquad \mathcal{D}_3 \to \hat{\theta}_3.$$

An **estimator** is a rule that converts data into an estimate.
For example, the sample mean estimates the unknown population mean:

$$\hat{\mu} = \bar{X}.$$



red cross: true value blue dots: estimates from repeated datasets

Bias and variance describe different errors

Suppose we repeat data collection many times and obtain estimates $\hat{\theta}$.

Bias

Measures systematic error:

$$\mathrm{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

A biased estimator is consistently shifted away from the truth.

Variance

Measures randomness across datasets:

$$\mathrm{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2].$$

A high-variance estimator changes a lot between datasets.

Bias describes the location of estimates; variance describes their spread.

Bias–variance decomposition

Mean squared error (MSE) measures the average squared distance between an estimate and the true parameter:

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2].$$

Adding and subtracting $\mathbb{E}[\hat{\theta}]$ gives

$$\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}[\hat{\theta}]) + (\mathbb{E}[\hat{\theta}] - \theta).$$

Taking expectations:

$$\boxed{\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2}.$$

An estimator can be inaccurate because it is biased, unstable, or both.

Example: the sample mean

For i.i.d. data with mean μ and variance σ^2,

$$\mathbb{E}[\bar{X}] = \frac{1}{n} \sum_i \mathbb{E}[X_i] = \mu.$$

Therefore, the sample mean is unbiased.
Independence gives

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n} \sum_i X_i \right) = \frac{\sigma^2}{n}.$$

Least-squares regression from likelihood

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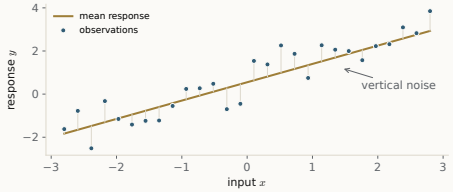
A probabilistic regression model

We model the target as a linear function of the input plus noise:

$$y_i = \boldsymbol{w}^\mathsf{T} \boldsymbol{x}_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

Here $\boldsymbol{x}_i$ is the input feature vector, $\boldsymbol{w}$ contains unknown parameters, and ϵ_i represents random error not explained by the model. Equivalently,

$$p(y_i \mid \boldsymbol{x}_i, \boldsymbol{w}) = \mathcal{N}(y_i \mid \boldsymbol{w}^\mathsf{T} \boldsymbol{x}_i, \sigma^2).$$



Why does Gaussian noise lead to squared error?

For independent observations,

$$-\ell(\boldsymbol{w}) = -\sum_i \log p(y_i \mid \boldsymbol{x}_i, \boldsymbol{w}).$$

Under Gaussian noise,

$$-\ell(\boldsymbol{w}) = \frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_i (y_i - \boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i)^2.$$

The first term does not depend on $\boldsymbol{w}$, so minimizing negative log-likelihood is equivalent to

$$\boxed{\hat{\boldsymbol{w}} = \arg\min_{\boldsymbol{w}} \sum_i (y_i - \boldsymbol{w}^\mathsf{T}\boldsymbol{x}_i)^2}.$$

Least squares is the maximum-likelihood solution under Gaussian noise.

Matrix form of least squares

Stack all samples into

$$\boldsymbol{X} \in \mathbb{R}^{n \times d}, \qquad \boldsymbol{y} \in \mathbb{R}^{n}.$$

The predictions for all samples are $\boldsymbol{X}\boldsymbol{w}$ and the residual vector is

$$\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}.$$

Therefore,

$$J(\boldsymbol{w}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2.$$

Least squares finds the parameters that make the residuals as small as possible.

Differentiating the least-squares objective

The least-squares objective is

$$J(\boldsymbol{w}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2.$$

Expanding the squared norm:

$$J = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w})^\mathsf{T}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}).$$

Expanding the terms gives

$$J = \boldsymbol{y}^\mathsf{T}\boldsymbol{y} - 2\boldsymbol{w}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{y} + \boldsymbol{w}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{w}.$$

Only terms containing $\boldsymbol{w}$ affect the gradient.

From gradient to normal equations

Differentiating each term with respect to $\boldsymbol{w}$:

$$\nabla_{\boldsymbol{w}} J = -2\boldsymbol{X}^{\mathsf{T}}\boldsymbol{y} + 2\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\boldsymbol{w}.$$

At the optimum,

$$\nabla_{\boldsymbol{w}} J = \mathbf{0}.$$

Therefore,

$$\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\hat{\boldsymbol{w}} = \boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}.$$

Equivalently,

$$\boldsymbol{X}^{\mathsf{T}}(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{w}}) = \mathbf{0}.$$

The residual is perpendicular to every feature direction.

Closed-form solution and practical computation

If $\boldsymbol{X}^\mathsf{T}\boldsymbol{X}$ is invertible,

$$\boxed{\hat{\boldsymbol{w}} = (\boldsymbol{X}^\mathsf{T}\boldsymbol{X})^{-1}\boldsymbol{X}^\mathsf{T}\boldsymbol{y}}.$$

This formula gives the mathematical solution, but practical implementations usually avoid explicitly computing the inverse.

- QR factorization;
- SVD;
- numerical least-squares solvers.

A minimal NumPy implementation

```
import numpy as np

# Solve least squares without forming (X.T X)^(-1)
w_hat, residuals, rank, singular_values = np.linalg.lstsq(
    X, y, rcond=None
)

print(rank, singular_values)
```

- Rank and singular values reveal numerical problems.
- Final evaluation should use held-out data.

Interpretation and summary

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Probability assumptions determine learning objectives

A probability model does more than describe data. It also determines how we measure prediction errors.
From this lecture:

$$\text{Gaussian noise} \implies -\log p(y|x, w) \implies \text{squared error.}$$

Other choices give other objectives:

Assumption	Resulting objective
Gaussian response noise	squared error
Bernoulli outcome	cross-entropy
Categorical outcome	cross-entropy

Why do model assumptions matter?

Maximum likelihood relies on assumptions about how data are generated. If these assumptions are unreasonable, the estimated parameters may not represent the real problem.
Examples:

- **Independence:** repeated measurements from the same patient may contain shared information.
- **Distribution choice:** Gaussian noise may not fit data with strong outliers.
- **Data coverage:** a model learned from one population may not generalize to another.

Summary

1. Likelihood evaluates how well a parameter explains observed data.
2. Maximum likelihood turns probability models into optimization objectives.
3. Gaussian assumptions lead to squared-error regression.
4. More data usually improve estimator stability, but assumptions matter.

Likelihood turns data into an estimation objective

Questions and discussion